

FORLI, Italy

WMO: 161470

Lat: 44.195N

Lon: 12.070E

Elev: 30

StdP: 100.97

Time Zone: 1.00 (EUC)

Period: 94-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.1	-2.8	-9.1	1.7	0.4	-7.2	2.1	1.0	8.0	9.1	6.7	6.9	1.5	210	0.404

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	34.1	22.8	32.8	22.5	31.1	21.8	24.1	31.3	23.2	30.4	22.4	29.4	3.2	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.0	16.7	27.6	21.0	15.7	26.8	20.0	14.8	26.1	72.5	31.5	69.0	30.5	66.0	29.1	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.4	6.2	5.3	-7.3	36.5	2.3	1.4	-8.9	37.5	-10.2	38.4	-11.5	39.2	-13.1	40.2		
			-8.1	25.6	2.1	1.1	-9.6	26.4	-10.9	27.0	-12.1	27.6	-13.6	28.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.2	4.1	5.3	9.6	12.7	17.7	21.9	24.3	24.0	19.7	15.2	10.0	4.9
	DBStd	7.75	2.81	3.19	3.38	2.88	2.92	3.04	2.45	2.78	2.80	3.27	3.53	3.19
	HDD10.0	578	182	132	48	9	0	0	0	0	0	3	44	159
	HDD18.3	2085	440	364	272	170	47	5	0	0	17	104	251	416
	CDD10.0	2104	1	2	34	90	240	358	443	435	291	166	42	2
	CDD18.3	571	0	0	0	1	28	113	185	177	58	8	0	0
	CDH23.3	5682	0	0	1	13	278	1128	1984	1807	426	46	0	0
	CDH26.7	2161	0	0	0	1	57	399	835	772	94	3	0	0
Wind	WSAvg	1.7	1.4	1.7	2.0	2.0	1.9	1.9	2.1	1.8	1.6	1.5	1.4	1.5
Precipitation	PrecAvg	660	36	45	54	62	53	45	34	49	74	78	76	53
	PrecMax	928	87	159	120	119	138	114	107	110	162	160	159	162
	PrecMin	430	1	2	6	20	10	14	2	3	24	14	13	7
	PrecStd	136	25	34	35	26	29	25	23	30	39	42	40	36
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.2	16.5	22.2	25.0	30.9	34.2	35.8	36.2	31.2	26.9	21.1	16.0
		MCWB	10.2	10.7	14.3	17.3	20.6	22.9	23.2	23.3	22.4	18.7	15.9	11.2
	2%	DB	12.9	14.8	19.8	22.2	28.0	32.3	34.0	34.4	29.1	24.8	18.8	13.2
		MCWB	9.0	9.7	13.0	15.2	19.2	22.4	22.8	23.0	20.9	18.0	14.2	9.8
	5%	DB	10.2	13.0	18.0	21.0	26.1	30.8	32.2	32.8	27.2	23.0	17.0	11.1
		MCWB	7.3	8.6	12.1	14.6	18.0	22.0	22.2	22.4	19.7	17.3	13.6	8.2
	10%	DB	8.9	11.2	16.2	19.2	24.7	29.0	31.0	30.9	25.9	21.0	15.4	9.9
		MCWB	6.5	7.7	11.2	13.3	17.3	20.6	21.6	21.8	18.7	16.2	12.7	7.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.0	12.1	15.3	17.8	21.4	24.8	25.1	26.0	23.5	20.2	17.1	12.4
		MCDB	14.0	15.4	20.5	23.6	27.9	31.2	32.6	33.6	29.2	24.1	19.4	14.6
	2%	WB	9.3	10.3	13.7	16.1	20.0	23.4	23.8	24.3	22.0	18.8	15.2	10.4
		MCDB	11.8	13.2	18.2	20.8	26.1	30.1	31.5	31.5	27.7	22.9	17.6	12.6
	5%	WB	7.9	9.3	12.7	15.2	18.9	22.3	23.0	23.1	20.8	17.8	14.1	9.1
		MCDB	9.7	12.1	17.0	19.5	24.6	29.2	30.5	30.3	25.9	21.9	16.4	10.8
	10%	WB	6.7	8.2	11.6	14.2	18.0	21.3	22.3	22.3	19.6	16.8	12.9	7.8
		MCDB	8.1	10.6	15.7	18.2	23.4	27.7	29.3	29.3	24.3	20.2	15.0	9.4
Mean Daily Temperature Range	5% DB	MDBR	7.4	9.6	10.7	10.6	11.2	11.4	11.9	11.8	10.7	9.0	7.5	7.4
		MCDBR	11.1	12.6	13.5	13.5	13.7	13.6	13.6	14.1	12.6	11.5	9.4	9.7
		MCWBR	8.0	8.4	8.1	7.8	6.8	6.1	6.1	6.4	6.3	6.1	6.4	6.9
	5% WB	MCDBR	9.4	11.0	12.2	11.7	12.4	12.6	12.4	12.6	11.6	10.4	7.8	7.9
		MCWBR	7.2	7.9	8.0	7.4	6.8	6.2	6.2	6.3	6.4	6.2	6.1	6.0
Clear-Sky Solar Irradiance	taub	0.353	0.374	0.410	0.442	0.442	0.451	0.442	0.435	0.422	0.411	0.378	0.350	
	taud	2.397	2.309	2.221	2.155	2.202	2.224	2.256	2.293	2.302	2.308	2.336	2.425	
	Ebn at Noon	754	801	818	821	833	825	828	820	798	748	710	721	
	Edh at Noon	78	100	124	143	141	138	133	124	113	99	82	70	
All-Sky Solar Radiation	RadAvg	1.37	2.22	3.49	4.64	5.78	6.58	6.75	5.78	4.21	2.64	1.52	1.21	
	RadStd	0.20	0.33	0.46	0.43	0.48	0.35	0.33	0.37	0.28	0.29	0.18	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (20 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	-88	+84	+45

Nomenclature: See separate page